

10 minutes

Biological Simulations

Stocks, Advice, and Scenarios

Advice, scenarios, and how CREST turns them into a resilience tool

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The thread (10 minutes)

- **Advice** is conditioned on the *past* — and can flip at a benchmark
- **Scenarios** ask “what if?” under fixed fishing — productivity, shocks, noise
- **CREST** carries those biological trajectories through to fleet value, effort and profit

Resilience means being ready for flips — the future will not be like the past.

Resilience in a Changing World

- **Can't Bounce Back** — productivity and hence reference points are *changing*
- **Stay Viable** — not restore an equilibrium

Why it matters

- Past strategies can fail under new regimes
- Fisheries: shifting productivity, distribution, recruitment, ...

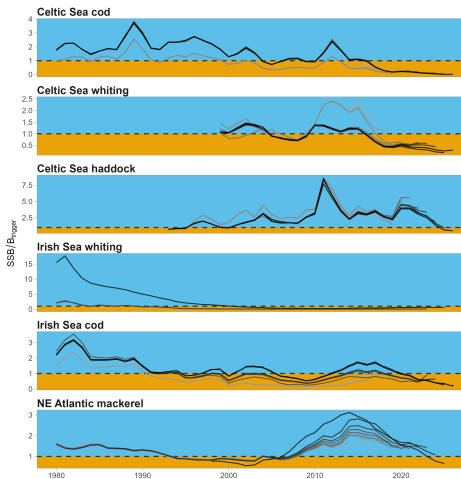
What resilience means

- **Persist** — keep functioning
- **Adapt** — change with conditions
- **Transform** — change the system when needed

1. Advice — status vs B_{trigger}

Trend and reference points:

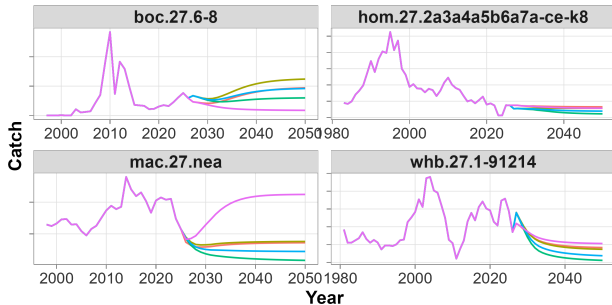
- $SSB/B_{\text{trigger}} = 1$?
 - > 1 — carry on
 - < 1 — reduce F , catch, effort
- Crossing the line flips status and advice
- A benchmark can change B_{trigger}



SSB/B_{trigger} — blue above 1, orange below

2. Scenarios — “What Ifs”

- Fixed fishing effort; examples not advice
- Status-quo; recruitment?
- Compare *within* a stock: the gap is the assumption that changed
- **Resilience**: under current fishing, does catch get better, hold, or erode?

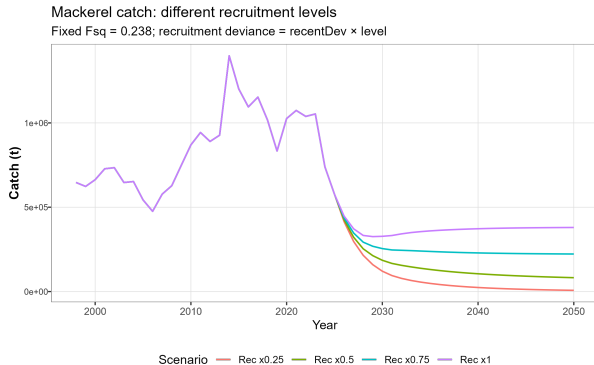


Pelagics — projected catch by scenario

3. Mackerel — Recruitment

Worked example at status-quo F :

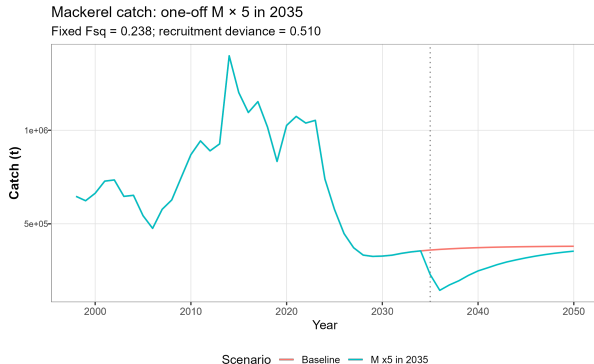
- Recruitment deviance $\times 1, 0.75, 0.5, 0.25$
- If effort stays as now, how does **catch** look?



Catch under recruitment multipliers

3. Mackerel — One-off M shock

- Same status-quo F/Effort
- Natural mortality $\times 5$ in **one year** (2035)
- A pulse — what a smooth forecast misses

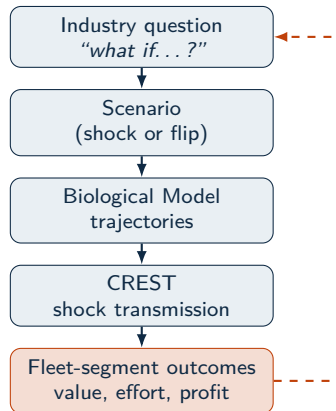


Catch: baseline vs $M \times 5$ in 2035

4. CREST — not an answering machine

CREST: Capacity, Resilience and Economic Shock Transmission

- Industry asks “*what if . . . ?*”
- Is the fleet resilient — or should it be?
- Biological shocks → value, effort, profit
- Between stock assessment and a whole-economy model



Takeaways

- Advice is based on the past — benchmarks can reclassify status
- Scenarios probed the future under advice rule — recruitment, shocks, noise
- Mackerel shows how catch responds to productivity and one-off flips
- CREST allows the fleet to ask the question

Full deck: [biological_simulations_slides.pdf](#)